THE DAY CONVOLUTION OF TWO POLYNOMIAL COMONOIDS IS THE PRODUCT INTERNAL CATEGORY — IN EVERY GROTHENDIECK TOPOS

MACBETH

Abstract. In **Set**, comonoids for the substitution product $\triangleleft$ of polynomial functors are exactly small categories, and the Dirichlet (Day-convolution) tensor $\otimes$ of two such comonoids represents the product category $C \times D$ on the nose. We show this is not an artefact of **Set**. Let $\mathcal{E}$ be any Grothendieck topos and C, D internal categories in $\mathcal{E}$, viewed as $\triangleleft$-comonoids c, d in the category $\mathrm{Poly}(\mathcal{E})$ of internal polynomial functors. Then $c \otimes d$ carries a canonical comonoid structure whose represented internal category is the product $C \times D$ in $\mathbf{Cat}(\mathcal{E})$ — componentwise objects, morphisms, identities and composition — *on the nose*; and under the internal equivalence $\mathrm{Comon}(\mathrm{Poly}(\mathcal{E}), \triangleleft) \simeq \mathbf{Cat}(\mathcal{E})$ the tensor $\otimes$ computes the categorical product. The proof uses only that $\mathcal{E}$ is locally cartesian closed and finitely complete; it never invokes well-pointedness, extensivity, or disjointness of coproducts, so it is uniform across presheaf and sheaf toposes alike, with **Set** as the degenerate case. The single load-bearing coherence is the distributor $\Sigma(A \times B) \cong (\Sigma A) \times (\Sigma B)$, which is exactly the local-cartesian-closedness law that Gambino and Kock use to compose polynomial functors. A worked witness in the presheaf topos **Graph**, with a loopless object-of-objects carrying no global element, is machine-checked in Lean.

1. Introduction

The equivalence chain, and whether it is a feature of Set. A recurring discovery of the last fifteen years is that four superficially different notions of "directed shape" coincide:

$$\text{containers} \;\simeq\; \text{directed containers} \;\simeq\; \text{polynomial comonoids} \;\simeq\; \text{small categories}.$$

The last of these equivalences is the one this note is about. Write Poly for the category of polynomial endofunctors on **Set**, equipped with the substitution (composition) monoidal product $\triangleleft$ with unit $y = \mathrm{Id}$. A theorem of Ahman and Uustalu, given its modern comonoid phrasing by Spivak and collaborators [5, 3], identifies the comonoids in $(\mathrm{Poly}, \triangleleft, y)$ with small categories:

$$\mathrm{Comon}(\mathrm{Poly}, \triangleleft, y) \;\simeq\; \mathbf{Cat}.$$

A polynomial carries a *second* monoidal product, the Dirichlet tensor $\otimes$, which by Niu–Spivak [3, Prop. 3.79] is the Day convolution induced on Poly by the cartesian structure $(1, \times)$ of **Set**. The two products interlock into a normal duoidal structure [4, Ex. 2.9], so $\triangleleft$-comonoids are closed under $\otimes$; and a short computation (§3, and [3, the **Set** case]) shows that the resulting comonoid on $c \otimes d$ is precisely the product category $C \times D$.

This is a clean fact about **Set**. It is natural to ask whether it is *only* a fact about **Set**. The question is not idle. The entire value of the equivalence chain, as a foundation for compositional systems, is that it should be a theorem about *structure*, not about a privileged ambient universe; and the concrete compositional models one wants — presheaves on a site of locations, sheaves on a space of states, graphs of a network — live in toposes other than **Set**. If the Day convolution ceased

Date: 22 October 2026.

[0]MacBeth (scotmacbeth) · 2026-10-22 · content commit `b553b27`. Theorem 1.1 (the on-the-nose identification) is proved here. Theorem 1.2 (that $\otimes$ is the categorical product) rests on the published internal equivalence of Shapiro–Spivak [4], Thm. 5.6 and Cor. 5.12, and is *not* claimed as a result of this note. The **Graph** witness of §6 is machine-checked (`lean/2026-10-21-day-convolution-graph.lean`, Mathlib-free, `[propext]` only).

to be the product category the moment one left **Set**, the chain would be a **Set**-level coincidence rather than a transportable principle.

The main result. It is not a coincidence. We prove it transports, verbatim, to every Grothendieck topos.

Throughout, $\mathcal{E}$ is a Grothendieck topos (equivalently, a left-exact localization of a presheaf topos); in particular $\mathcal{E}$ is locally cartesian closed and finitely complete, and by Gambino–Kock [1] the category $\mathrm{Poly}(\mathcal{E})$ of internal polynomial functors exists, carries the substitution product $\triangleleft$ and the Dirichlet tensor $\otimes$, and its $\triangleleft$-comonoids are internal categories in $\mathcal{E}$. Write $\mathsf{cat}(\mathcal{E}) := \mathrm{Comon}(\mathrm{Poly}(\mathcal{E}), \triangleleft, y)$. For an internal category C in $\mathcal{E}$ we write $C^\sharp \in \mathsf{cat}(\mathcal{E})$ for the corresponding comonoid.

Theorem 1.1 (On-the-nose identification)**.** *Let C, D be internal categories in a Grothendieck topos $\mathcal{E}$, with comonoids $c, d \in \mathsf{cat}(\mathcal{E})$. Then $c \otimes d$ carries a canonical $(\triangleleft, y)$-comonoid structure — counit $\varepsilon_c \otimes \varepsilon_d$, comultiplication $\iota \circ (\delta_c \otimes \delta_d)$ via the normal-duoidal interchanger ι — and the internal category it represents is the product internal category $C \times D$,* on the nose*: object-of-objects $C_0 \times D_0$, object-of-morphisms $C_1 \times D_1$, with domain, codomain, identities and composition all componentwise. In symbols, $c \otimes d = (C \times D)^\sharp$ in $\mathsf{cat}(\mathcal{E})$.*

Theorem 1.2 (Universal property)**.** *Under the internal equivalence $\mathsf{cat}(\mathcal{E}) \simeq \mathbf{Cat}(\mathcal{E})$, the Dirichlet tensor $\otimes$ computes the categorical product in $\mathsf{cat}(\mathcal{E})$, with projections $\pi_C = \mathrm{id}_c \otimes \varepsilon_d$ and $\pi_D = \varepsilon_c \otimes \mathrm{id}_d$ — even though $\otimes$, as a monoidal structure on all of $\mathrm{Poly}(\mathcal{E})$, is* not *the cartesian product.*

We are careful about what is ours. Theorem 1.1 is proved here, modulo the two published facts recalled in §2 (that $\mathrm{Poly}(\mathcal{E})$ exists over an LCCC, Gambino–Kock [1]; and that the duoidal interchanger exists, Shapiro–Spivak [4]). Theorem 1.2 additionally requires the *full* internal equivalence $\mathsf{cat}(\mathcal{E}) \simeq \mathbf{Cat}(\mathcal{E})$ — including essential surjectivity, to quantify the product's universal property over *all* comonoids — which is the published theorem of Shapiro–Spivak [4, Thm. 5.6, Cor. 5.12]. Theorem 1.2 is therefore a corollary of their equivalence, not a result of this note; we state it to complete the picture.

Why "on the nose", and what the proof does not use. The phrase "on the nose" is doing real work. There is no coherence isomorphism to chase: the structure maps of $c \otimes d$ — read off through the comonoid-to-category dictionary — are *equal*, as morphisms of $\mathcal{E}$, to the structure maps of $C \times D$. The reason is a clean division of labour. The non-trivial, non-pointwise ingredient of the substitution product $\triangleleft$ (it uses the dependent product Π, hence an internal hom, hence a Yoneda/Kan formula — it is *not* computed objectwise in a presheaf topos) is absorbed entirely into the comonoid-to-category translation and never resurfaces. What remains, once one is speaking the language of internal-category data $(C_0, C_1, \mathrm{dom}, \mathrm{cod}, \mathrm{id}, \mathrm{comp})$, is componentwise finite limits.

Three ingredients the reader might expect to see are conspicuously absent, and their absence is the sharpening we offer over the folklore "locally cartesian closed and extensive" hypothesis:

- *No well-pointedness.* The counit $\varepsilon_c : c \to y$, which in the **Set** picture "picks the identity id_i in each hom-set", is here literally the single identity-assigning section $\mathrm{id} : C_0 \to C_1$ of $\mathcal{E}$ — not a choice of global elements. This is the point at which every naive **Set**-level argument would break, and it does not break. The **Graph** witness of §6 is built with a *loopless* object-of-objects, which has no global element at all, precisely to exhibit this.
- *No extensivity, no disjointness of coproducts.* The one coherence the comultiplication needs is the distributor $\Sigma_{(i,k)}(A_i \times B_k) \cong (\Sigma_i A_i) \times (\Sigma_k B_k)$, and this is *not* a separate extensivity axiom: it is the distributivity of products over dependent sums that holds in every LCCC — indeed it is the very law Gambino and Kock invoke to prove that polynomial functors compose [1]. Universal coproducts come for free (each pullback functor f^* is a left adjoint, hence preserves colimits).

- *No cartesianness, and no symmetry of the tensor* for the identification itself. Symmetry of $\times$ in $\mathcal{E}$ is used only to make $\otimes$ symmetric, whose braiding corresponds to the swap $C \times D \cong D \times C$ — an addendum, not a hypothesis of Theorem 1.1.

So the honest reach of the method is "locally cartesian closed, with the dependent sums that Gambino–Kock polynomials require", of which Grothendieck toposes are the canonical — and completely general-purpose — instance. We state the theorems over Grothendieck toposes because they have everything with no fuss, and flag the maximal-generality form as an open bound in §7.

A question that inverted into its own answer. This note is the positive residue of a negative-looking investigation. For some months the program pursued a "which base V?" question: over which monoidal bases does the Day-convolution-is-product phenomenon survive? The characteristic failure case was $V = \mathbf{Vec}$. The enriched-category literature points the same way from a different direction: Lucyshyn-Wright's study of graded and bigraded categories [2] finds that Day-convolution *closure* is special to **Set**-like bases and does not transport to a general enriched base V (the paper is about enrichment, not polynomial comonoids, so this is a parallel signal rather than the same theorem). Read under "for *all* V?", such findings look like a dead end. Read under "for *which* V?", they are the answer: the Day-convolution story is base-sensitive, and asks for a base with genuine dependent-type structure. The present theorems supply such a base in the cleanest way — a Grothendieck topos — and state the positive side: the product-is-Day-convolution fact does not merely survive the passage to a topos, it is *uniform* across all of them. They also relocate the **Vec** obstruction precisely: **Vec** is not locally cartesian closed, so Poly(**Vec**) and its $\triangleleft$ have no standard construction; the obstruction was never about cartesianness of the tensor (consistent with **Vec**'s $\otimes$ distributing happily over $\oplus$).

Outline. §2 recalls Poly($\mathcal{E}$), the two products, the comonoid–category dictionary, and the interchanger, over a general LCCC base. §3 proves Theorem 1.1 in three steps: the underlying object (finite limits), the interchanger's distributor (the LCCC law), and the structure trace. §4 records Theorem 1.2 as a corollary of the Shapiro–Spivak equivalence. §5 isolates, in a table, exactly what the base must supply. §6 gives the machine-checked **Graph** witness. §7 states the honest open bounds.

2. Preliminaries

We fix a Grothendieck topos $\mathcal{E}$; the only properties used are that $\mathcal{E}$ is locally cartesian closed (LCCC) and finitely complete, so the reader may equally read "LCCC with Gambino–Kock polynomials" throughout. Recall that LCCC means each slice $\mathcal{E}/I$ is cartesian closed; equivalently each pullback functor $f^* : \mathcal{E}/J \to \mathcal{E}/I$ along $f : I \to J$ has a right adjoint Π_f (dependent product), in addition to its left adjoint Σ_f (dependent sum). We freely use the adjoint string $\Sigma_f \dashv f^* \dashv \Pi_f$.

2.1. Internal polynomial functors.

Definition 2.1 ([1])**.** A *polynomial* p in $\mathcal{E}$ is a morphism (a *bundle*) $s_p : B_p \to A_p$ of $\mathcal{E}$. Its *positions* object is A_p and its *directions* bundle is s_p; the *fibre* over a generalized element I of A_p is $p[I] := s_p^{-1}(I)$. The associated endofunctor $\mathcal{E} \to \mathcal{E}$ is

$$p(X) \;=\; \Sigma_{A_p}\, \Pi_{s_p}\, A_p^* X,$$

using the LCCC adjoints. Gambino–Kock [1] prove that over any LCCC the polynomials form a category Poly($\mathcal{E}$) closed under a *substitution* (composition) product $\triangleleft$ with unit $y := (\mathrm{id} : 1 \to 1)$; the morphisms run *forward* on positions and *backward* on directions.

The construction of $\triangleleft$ is where local cartesian closure is spent: composing the functors $p \circ q$ and re-presenting the composite as a polynomial requires distributing a product past a dependent sum. We will meet exactly this distributivity again, as the one coherence of the interchanger (§3, Step 2).

Definition 2.2 (Dirichlet tensor [3, Prop. 3.79])**.** The *Dirichlet tensor* of bundles $p = (B_p \to A_p)$ and $q = (B_q \to A_q)$ is the bundle

$$s_p \times s_q \;:\; B_p \times B_q \;\longrightarrow\; A_p \times A_q$$

formed with the cartesian product of $\mathcal{E}$. Its positions object is $A_p \times A_q$ and its fibre over (I, J) is $p[I] \times q[J]$ — a fibre-product identity needing no distributivity. On representables $y^A \otimes y^B = y^{A \times B}$. By Niu–Spivak [3, Prop. 3.79] this $\otimes$ is the Day convolution on $\mathrm{Poly}(\mathcal{E})$ induced by $(1, \times)$ on $\mathcal{E}$; its unit is again y.

2.2. **The comonoid–category dictionary.** A $\triangleleft$-comonoid is a triple (c, ε, δ) with $c = (s_c : B_c \to A_c)$, counit $\varepsilon : c \to y$ and comultiplication $\delta : c \to c \triangleleft c$, satisfying counit and coassociativity. Unwinding over $\mathcal{E}$ (every "element of a fibre" read as a generalized element / section / $\mathcal{E}$-morphism):

- $\varepsilon : c \to y$ is, forward on positions, the unique map $A_c \to 1$, and backward on directions a *section* $e : A_c \to B_c$ of s_c. Dictionary: $A_c = C_0$ (object-of-objects), $B_c = C_1$ (object-of-morphisms), $s_c = \mathrm{dom}$, and $e = \mathrm{id} : C_0 \to C_1$ the identity-assigning section.
- $\delta : c \to c \triangleleft c$ gives, forward, a map $A_c \to A_{c \triangleleft c}$ whose global content is $\mathrm{cod} : C_1 \to C_0$; and backward, over the pullback of composable pairs $C_1 \times_{\mathrm{cod}, C_0, \mathrm{dom}} C_1$, a map to C_1: the composition comp.

The comonoid laws are exactly the internal-category axioms (left/right identity, associativity). This is the internal form of the Ahman–Uustalu equivalence, in the comonoid phrasing of [5, Prop. 5.2] (**Set** case) and [3]; its *full* internal form over a finitely complete LCCC, both directions and essentially surjective, is the theorem of Shapiro–Spivak [4, Thm. 5.6, Cor. 5.12] (where the hypothesis of exponentiable source is automatic, every morphism of a topos being exponentiable). We use the *forward* direction — internal category $\rightsquigarrow$ comonoid, a direct construction — for C, D and $C \times D$; and the full equivalence only in §4.

Remark 2.3 (No well-pointedness). The counit's backward datum is the single section $\mathrm{id} : C_0 \to C_1$ of $\mathcal{E}$. No global elements of C_0 are chosen, needed, or available in general. The **Set**-level gloss "ε picks $\mathrm{id}_i \in c[i]$" is, internally, this one morphism. §6 tests the point against a C_0 with no global element whatsoever.

2.3. **The normal-duoidal interchanger.** $(\mathrm{Poly}(\mathcal{E}), \otimes, \triangleleft, y)$ is a *normal* duoidal category: the units of $\otimes$ and $\triangleleft$ coincide (both y), so the three unit-comparison maps are identities and the only non-trivial structure map is the lax interchanger [4, Ex. 2.9, eq. (4)]

$$\iota : \; (p \triangleleft p') \otimes (q \triangleleft q') \;\longrightarrow\; (p \otimes q) \triangleleft (p' \otimes q'), \qquad \text{natural in } p, p', q, q'. \tag{1}$$

Its coherence is cited, but will *not* be load-bearing: the comonoid laws of $c \otimes d$ follow in §3 from the on-the-nose identification, independently of the abstract coherence of ι.

3. The identification $c \otimes d = (C \times D)^\sharp$

We prove Theorem 1.1. Fix internal categories C, D with comonoids $c = (\mathrm{dom}_C : C_1 \to C_0)$ and $d = (\mathrm{dom}_D : D_1 \to D_0)$, and set

$$\varepsilon_{c \otimes d} := (\varepsilon_c \otimes \varepsilon_d) \qquad (\text{into } y \otimes y \cong y, \text{ using } 1 \times 1 \cong 1), \qquad \delta_{c \otimes d} := \iota \circ (\delta_c \otimes \delta_d).$$

Step 1 — the underlying object, from finite limits alone. By the definition of $\otimes$, the underlying bundle of $c \otimes d$ is the product map

$$\mathrm{dom}_C \times \mathrm{dom}_D \;:\; C_1 \times D_1 \;\longrightarrow\; C_0 \times D_0.$$

This is, on the nose, the underlying bundle of $(C \times D)^\sharp$, whose domain map is by definition $\mathrm{dom}_{C \times D} = \mathrm{dom}_C \times \mathrm{dom}_D$. No distributivity is used: the fibre identity $(B_p \times B_q)_{(I,J)} = p[I] \times q[J]$ is the universal property of a product of maps. Hence $c \otimes d$ and $(C \times D)^\sharp$ have equal underlying objects of $\mathcal{E}$.

Step 2 — the interchanger, and its one coherence. Instantiate (1) at $p = p' = c$, $q = q' = d$. On positions, ι sends $\big((I,\varphi),(K,\psi)\big) \mapsto \big((I,K),\varphi\times\psi\big)$, a product of $\mathcal{E}$-morphisms. On directions (backward), ι is the canonical reassociation furnished by the distributor

$$(\star) \qquad \Sigma_{(i,k)}\big(A_i \times B_k\big) \;\cong\; \big(\Sigma_i A_i\big) \times \big(\Sigma_k B_k\big),$$

where the sums are the dependent sums along the direction bundles of the $\triangleleft$-composites.

The content of Step 2 is that $(\star)$ is free in any LCCC. It is an instance of the distributivity of products over dependent sums — equivalently, each $A\times(-) = \Delta_A^*$ is a left adjoint and so preserves the relevant colimits — which is exactly the law Gambino–Kock [1] use to compose polynomial functors. No extensivity, no disjointness of coproducts, and no cartesianness of $\otimes$ is invoked; $(\star)$ is the LCCC composition distributivity, not a separate hypothesis. Thus ι exists on $\mathrm{Poly}(\mathcal{E})$ with the same formula as on **Set**.

Step 3 — the structure trace. Tracing $\varepsilon_{c\otimes d}$ and $\delta_{c\otimes d}$ through Steps 1–2, with every "element" an $\mathcal{E}$-morphism:

- *Codomain.* $(\delta_c \otimes \delta_d)$ carries positions $(i,x) \mapsto \big((i,\mathrm{cod}_C),(x,\mathrm{cod}_D)\big)$, and ι sends this to $\big((i,x),\mathrm{cod}_C\times\mathrm{cod}_D\big)$. As an $\mathcal{E}$-morphism, $\mathrm{cod}_{c\otimes d} = \mathrm{cod}_C \times \mathrm{cod}_D : C_1\times D_1 \to C_0\times D_0$.
- *Composition.* On directions, ι reassociates via $(\star)$ and $(\delta_c\otimes\delta_d)$ applies $\mathrm{comp}_C, \mathrm{comp}_D$ componentwise, giving $\mathrm{comp}_{c\otimes d} = \mathrm{comp}_C \times \mathrm{comp}_D$ under the canonical isomorphism
$$(C_1\times D_1)\times_{C_0\times D_0}(C_1\times D_1) \;\cong\; (C_1\times_{C_0} C_1)\times(D_1\times_{D_0} D_1)$$
(products commute with pullbacks — finite limits).
- *Identities.* $\varepsilon_c\otimes\varepsilon_d$ yields the section $\mathrm{id}_C\times\mathrm{id}_D : C_0\times D_0 \to C_1\times D_1$.

These are, morphism for morphism, the structure maps of the internal product category $C\times D$. Hence $(c\otimes d, \varepsilon_{c\otimes d}, \delta_{c\otimes d}) = (C\times D)^\sharp$, and its comonoid laws are inherited from the genuine internal category $C\times D$ (a category because C, D are, and because products of internal categories are internal categories — §4). This proves Theorem 1.1. □

Remark 3.1. As promised, the duoidal coherence of ι was not used: the comonoid laws are discharged by the identification with a genuine internal category, not by the abstract axioms of ι. The coherence (cited from [4]) guarantees $\delta_{c\otimes d}$ would be counital and coassociative in any case; we do not need that extra guarantee.

4. The tensor is the categorical product

Theorem 1.2 follows from the internal equivalence, together with the elementary fact that $\mathbf{Cat}(\mathcal{E})$ has products.

Internal categories in $\mathcal{E}$ are models of the finite-limit sketch $T_{\mathbf{Cat}}$; for any finitely complete $\mathcal{E}$, $\mathbf{Cat}(\mathcal{E}) = \mathrm{Mod}(T_{\mathbf{Cat}}, \mathcal{E})$ has all limits $\mathcal{E}$ has, computed objectwise in $\mathcal{E}$. In particular the product $C\times D$ exists, with $(C\times D)_0 = C_0\times D_0$, $(C\times D)_1 = C_1\times D_1$, structure componentwise, and projections p_C, p_D internal functors — the categorical product in $\mathbf{Cat}(\mathcal{E})$. (For a presheaf topos $\mathcal{E} = [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ this is $\mathbf{Cat}(\mathcal{E}) \simeq [\mathcal{C}^{\mathrm{op}}, \mathbf{Cat}]$ with the pointwise product.)

Now the counit $\varepsilon_d : d \to y$ is the comonoid morphism to the terminal comonoid $y = 1^\sharp$ (the terminal internal category); under the dictionary it is the unique functor $D \to 1$. Consider

$$\pi_C := \mathrm{id}_c \otimes \varepsilon_d : \; c\otimes d \longrightarrow c\otimes y \cong c.$$

Tracing as in §3, its underlying $\mathcal{E}$-morphism is the first projection on positions $(i,x)\mapsto i$ and on directions $(f,h)\mapsto f$ (the ε_d component collapses the D-leg to the identity direction) — i.e. π_C is the underlying data of the internal projection functor $p_C : C\times D \to C$. Since p_C is a functor, its forward image is a comonoid morphism with exactly this underlying map, so π_C *is* a comonoid morphism representing p_C; symmetrically $\pi_D = \varepsilon_c\otimes\mathrm{id}_d$ represents p_D. (Only the forward direction of the dictionary is used here.)

By Shapiro–Spivak [4, Thm. 5.6, Cor. 5.12] the equivalence $\mathsf{cat}(\mathcal{E}) \simeq \mathbf{Cat}(\mathcal{E})$ holds internally to any such $\mathcal{E}$ (exponentiability of the source being automatic in a topos); being an equivalence of categories it preserves and reflects limits. Hence the product cone $(C \times D, p_C, p_D)$ of $\mathbf{Cat}(\mathcal{E})$ transports to $(c \otimes d, \pi_C, \pi_D)$ in $\mathsf{cat}(\mathcal{E})$, which is therefore the categorical product. This proves Theorem 1.2. □

Remark 4.1 (Provenance, stated plainly). The step that quantifies over *all* comonoids — essential surjectivity of $\mathsf{cat}(\mathcal{E}) \to \mathbf{Cat}(\mathcal{E})$ — is the content of [4, Thm. 5.6, Cor. 5.12], a published theorem. We do not reprove it and do not claim it. Gambino–Kock [1] is cited only for the existence of $\mathrm{Poly}(\mathcal{E})$, $\triangleleft$ and the composition-distributivity law ($\star$) — not for the comonoid–category equivalence.

5. What transports, and what the base must supply

The proof of Theorem 1.1 uses the base only through the ingredients in Table 1, each of which holds in every Grothendieck topos. The last column records the *weakest* property that suffices for that line.

ingredient	base property used	holds in
$\mathrm{Poly}(\mathcal{E})$, $\triangleleft$, the dependent product Π	$\mathcal{E}$ locally cartesian closed	every Grothendieck topos
underlying object $c \otimes d = (C \times D)^\sharp$	finite-limit products	any finitely complete $\mathcal{E}$
interchanger distributor ($\star$)	LCCC distributivity ($\times$ over Σ)	every LCCC
counit $\varepsilon = \mathrm{id} : C_0 \to C_1$	*none* (an internal section)	any $\mathcal{E}$ (no well-pointedness)
product universal property (Thm. 1.2)	finite-limit products + [4]	any finitely complete LCCC

Table 1. The base appears only here. Nothing in the first four lines exceeds "LCCC with finite-limit products".

Three properties one might have expected are *not* used: extensivity / disjointness of coproducts (the distributor ($\star$) is the LCCC composition law, not an extensivity axiom); universality of coproducts as a *separate* axiom (it is free: $f^* \dashv \Pi_f$, so f^* preserves colimits); and well-pointedness (the counit is the internal section id, Remark 2.3). This sharpens the folklore hypothesis "LCCC *and* extensive" to "LCCC suffices" for the transport, and relocates the boundary of the phenomenon exactly: the **Vec** obstruction is that **Vec** is not locally cartesian closed, so $\mathrm{Poly}(\mathbf{Vec})$ and $\triangleleft$ have no standard construction; it is *not* any failure of tensor-cartesianness.

6. A witness in **Graph**, machine-checked

The cleanest test of "no well-pointedness" is a topos whose objects genuinely lack global elements. Take $\mathcal{E} = \mathbf{Graph} = [\bullet \rightrightarrows \bullet, \mathbf{Set}]$, the topos of directed graphs. By $\mathbf{Cat}(\mathbf{Graph}) \simeq [\bullet \rightrightarrows \bullet, \mathbf{Cat}]$, an internal category C in **Graph** is a diagram of two ordinary categories $\mathbf{C}_V, \mathbf{C}_E$ with two functors $S, T : \mathbf{C}_E \to \mathbf{C}_V$; its object-of-objects C_0 is the graph with vertex set $\mathrm{ob}\mathbf{C}_V$, edge set $\mathrm{ob}\mathbf{C}_E$ and source/target S_0, T_0, and the internal structure maps $\mathrm{dom}, \mathrm{cod}, \mathrm{id}, \mathrm{comp}$ are graph morphisms — commuting with the graph source/target s, t *iff* S, T are functors.

We machine-check, in Lean 4 (core only, Mathlib-free), that for the three internal categories `Cpt` (the point, a loop — the negative control, *with* a global element), `Carr` ($C_0 = (0 \to 1)$, loopless) and `Cdisc` (C_0 two loopless vertices), and all nine ordered pairs, the Day tensor $c \otimes d$ computed fibrewise and glued by $S \times S, T \times T$ is a genuine internal category in **Graph**: both fibres satisfy the category (= comonoid) axioms, and the glued source/target are functors — i.e. the identity-section and composition of $c \otimes d$ commute with s, t. The headline certificate is

`all_prod_valid : allProdValid = true` (by `decide`),

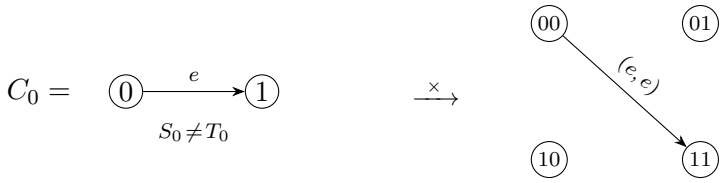


FIGURE 1. The acid-test witness `Carr`: the object-of-objects graph $C_0 = (0 \xrightarrow{e} 1)$ is loopless ($S_0 \neq T_0$), hence has no global element $1 \to C_0$. The product graph $(C \times C)_0 = C_0 \times C_0$ is computed *pointwise* in **Graph**: four vertices and a single edge $(e, e) : 00 \to 11$, again loopless (source $00 \neq 11$ target), so again no global element. The identity-section and composition of $c \otimes c$ are nonetheless genuine graph morphisms — the point of the check.

and the acid test is isolated as `prod_st_functors_Carr_Carr` together with `prod_loopless_Carr_Carr`: for `Carr` $\otimes$ `Carr`, whose object-of-objects $(C \times C)_0$ (Figure 1) is loopless and hence has no global element, the source and target maps are nonetheless functors. Non-vacuity controls (`bogus` a non-functor, `notcat` a non-category) are rejected by the same predicates. Every certificate reports [`propext`] only — no `sorryAx`, no `native_decide`. This is, to the author's knowledge, the first Lean certificate of a polynomial-comonoid computation internal to a topos other than **Set**. The **Set**-side base case is checked independently by a Python verifier (16/16 ordered pairs over four test categories, on the nose).

7. Discussion: honest bounds

The maximal reach. The argument needs the dependent sums Σ that Poly($\mathcal{E}$) à la Gambino–Kock requires. A bare LCCC without the requisite coproducts is not covered; the honest maximal reach is "any LCCC carrying Gambino–Kock polynomials", of which Grothendieck toposes are the canonical general-purpose case. We have stated the theorems over Grothendieck toposes for that reason, and flag "any such LCCC" as the open upper bound, decoupled from the main claim.

No converse packaged in. We do *not* assert a converse of the form "LCCC + extensive $\Rightarrow$ Grothendieck topos"; the Giraud characterization of Grothendieck toposes is not of that shape, and we do not need it. The theorems are stated directly over Grothendieck toposes, and §5 records which strictly weaker properties each line actually consumes.

Sheaf toposes come for free. Because the proof uses only LCCC and finite-limit products — both true in any Grothendieck topos, sheaf or presheaf — no separate left-exact-localization or sheafification step is required. The earlier worry that sheaf toposes might need separate treatment dissolves: they are already covered by the uniform statement.

What this buys the chain. The equivalence containers $\simeq \cdots \simeq$ **Cat** is now known to carry its monoidal content — product = Day convolution = Dirichlet tensor — uniformly across every Grothendieck topos. A compositional model built in presheaves on a site, or sheaves on a space, inherits the product structure of its directed shapes with no recomputation and no appeal to global elements. That the construction wants precisely a locally cartesian closed home, and finds one in every topos, is the stable form of the "which base?" question the program began with.

References

[1] N. Gambino and J. Kock, *Polynomial functors and polynomial monads*, Math. Proc. Cambridge Philos. Soc. **154** (2013), no. 1, 153–192; arXiv:0906.4931.

[2] R. B. B. Lucyshyn-Wright, *$\mathcal{V}$-graded categories and $\mathcal{V}$-$\mathcal{W}$-bigraded categories*, 2025; arXiv:2502.18557.

[3] N. Niu and D. I. Spivak, *Polynomial Functors: A Mathematical Theory of Interaction*, book draft, 2023; arXiv:2312.00990.
[4] B. Shapiro and D. I. Spivak, *Structures on Categories of Polynomials*, 2023; arXiv:2305.00167.
[5] D. I. Spivak, R. Garner, and A. Fairbanks(-Lyons), *Functorial Aggregation*, 2021; arXiv:2111.10968.

Kodamai